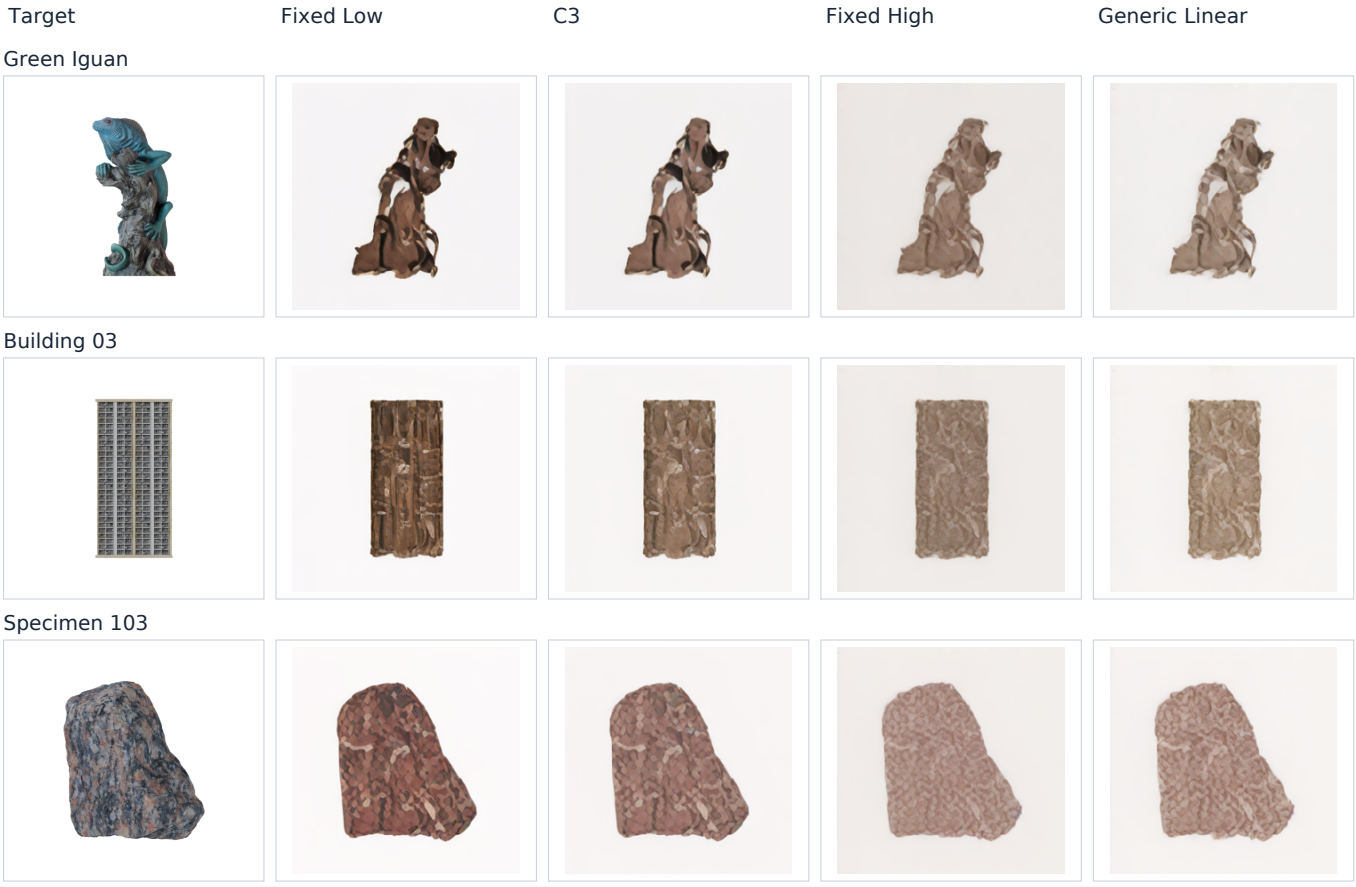


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. The C3–Fixed Low FG-PSNR differences for Building 03 and Specimen 103 are +2.644 **and** −0.468 **dB, respectively. Source attribution is in Supplementary Section S9.**



Fig. 3. GeoTex Bake10 appearance at held-out camera 004 for Stone. The row shows a GT-256 reference self-bake, Fixed Mean, and Late-high under a shared mesh/UV and within-object crop. The object has the lowest absolute eleven-view FG-LPIPS among seven display-cleared assets, while the Late-high minus Fixed Mean difference is −0.000464 **FG-LPIPS and** +0.001 **dB FG-PSNR. Source images are 512×512 and shown without method-specific color edits. The ten-object effect distribution and full credits are in Supplementary Section S5.**

4.4.3. Protocol and deployment boundaries

Separate twenty-object GSO evaluation. The GSO/MV-Adapter study completes twelve conditions by three seeds on twenty fixed inputs, plus twenty matched GT self-bakes: 740 unique baked meshes and 8,140 unseen-view records. The objects, source meshes, projection cameras, texture resolution and eleven unseen cameras are shared across controls; unseen cameras do not contribute to baking.



Fig. 3. GeoTex Bake10 appearance at held-out camera 004. Each row shows a GT-256 reference self-bake, Fixed Mean, and Late-high under a shared mesh/UV and within-object crop. Source images are 512×512 and displayed without method-specific color edits. Full source credits and the ten-object effect distribution are in Supplementary Section S5.

quantities contextualize the method effects and do not define a theoretical quality bound.

Native controls and protocol sensitivity. The added Native residual and No Adapter rows are post-inspection exploratory controls; their wrapper semantics and complete fifteen-test family are in Supplementary Section S5.

A separate, previously studied fifty-object checkpoint comparison gives Late-high–Fixed Mean changes of −1.184 dB nonreference PSNR and +0.00266 nonreference LPIPS; C3 has no corrected advantage over Fixed Mean or Fixed High. The changed checkpoint and protocol make this a sensitivity boundary, not an independent backbone confirmation (Supplementary Section S8).

Native-pipeline quality boundary. On the same fifty inputs, the new geometry checkpoint trails full v29 by 2.16 dB FG-PSNR and has FG-LPIPS 0.0341 higher. This matched-input deployment comparison changes base weights and geometry implementation and uses a nondefault 50-step protocol, so it does not isolate TCAS. A same-base follow-up remains 0.0159 LPIPS worse than full v29 despite a local unscaled-versus-disabled geometry benefit. Neither result establishes TCAS deployment superiority; full tests and protocol details are in Supplementary Section S8.

4.5. Sensitivity Across Native Conditioning Interfaces

The auxiliary studies ask related scheduling questions under different native conditioning operators and camera contracts. The 50-object and GSO100 studies both use MV-Adapter; GSO100 is a separate source-ordered cohort and protocol sensitivity, not a third independent backbone. MVDiffusion scales a partial correspondence operator and is reported only as additional context. Table 7 presents each study's own specified primary contrast: the GeoTex contrast compares Middle-high with the mean of Early-high and Late-high, whereas both MV-Adapter contrasts compare Middle-high with Fixed Mean. We report these rows separately and do not pool them or treat them as an identical replication.

Input-selected native MV-Adapter study. We selected fifty Fresh C inputs not previously evaluated through MV-Adapter, using input availability and native material/geometry compatibility. Native mesh normalization and six orthographic cameras define both generation and evaluation targets. Three-object identity checks confirmed agreement between the deployed and recorded default paths. An earlier seven-object panel had an invalid camera calibration and is excluded from these comparisons (Supplementary Section S3).

The seven conditions share object inputs, seed, geometry and evaluation code. With 50 steps, low/high are 0.75/1.0; stage-high schedules contain 16 high steps, Fixed Mean is 0.83, and Generic Linear is 0.75–0.91. These five controls have matched per-feature sums within this interface. All 50 objects remain in the primary cohort, including thirteen texture-absent sources with legal constant-white base-color targets. Targets are unlit native-camera base color, not Fresh C's differently normalized